

Mobile Apps Development

(Elective Course)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

1. To provide students with the knowledge of recent trends in mobile application development.
2. To give the overview of all existing mobile operating systems and the development SDKs required to develop applications on them.
3. To teach students the basics of application development in Android in depth.

1. Introduction to Mobile OSes

- 1.1 Introduction of Mobile OSes: Android, iOS, Ubuntu, Touch, Blackberry, Tizen, Firefox OS, Symbian, Windows Phone
- 1.2 Build and Structures of Mobile OSes
- 1.3. Introduction to development environment (Native v/s HTML5)
- 1.4. Introduction to Android
 - 1.4.1 API levels/versions of Android
 - 1.4.2. Pros and Cons of Android
 - 1.4.3. Comparison of Android with other Mobile OSes
- 1.5. Introduction to Android VM and Runtime (Dalvik and ART)
- 1.6. Installation and configuration of Android SDKs and Eclipse IDE
 - 1.6.1. Their integration using ADT Plugin
 - 1.6.2. Running an emulator
- 1.7. Using ADB command line interface

2. Android Classes and Basics

- 2.1. Android Fundamentals
 - 2.1.1. Creating an Android App,
 - 2.1.2. Android Manifests File
- 2.2 The Activity Class
 - 2.2.1. Activity Lifecycle
 - 2.2.2. Extending the activity class
 - 2.2.3. Creating Default Activity
 - 2.2.4. Creating Splash and Login Activities
- 2.3. The Intent Class
 - 2.3.1. Creating Intent
 - 2.3.2. Switching between Activities using Intent
- 2.4. Permissions
 - 2.4.1. Allow APP permissions in Android Manifest
- 2.5. The Fragment Class and Its usage

3. **Android User Interface**
 - 3.1. Introduction to Multiple Screen Size and Orientation Interfaces
 - 3.2. User Interface Classes
 - 3.3. Android XML Layouts, Resources and Styles
 - 3.4. Android 3rd party UI/UX Libraries
4. **Advanced Topics**
 - 4.1. User Notifications
 - 4.2. The BroadcastReceiver class
 - 4.3. Threads, AsyncTask and Handlers
 - 4.4. Alarms
 - 4.5. Networking
5. **Graphics and Multimedia**
 - 5.1. Graphics and Animations
 - 5.2. Multitouch and Gestures
 - 5.3. Multimedia
6. **APIs and Sensors**
 - 6.1. Sensors
 - 6.2. Camera API
 - 6.3. Location and Maps API
 - 6.4. Device Manager API
 - 6.5. Play Services API
 - 6.6. Multiplayer Gaming API
7. **Packaging and Monetizing**
 - 7.1. Data Management (using sqlite database, local storage)
 - 7.2. The ContentProvider Class
 - 7.3. The Service Class
 - 7.4. Google Mobile Ads SDK
 - 7.5. Signing and Exporting an APP
 - 7.6. Publishing your app to the Play Store

Laboratory:

The laboratory classes should be based on the topics covered in the lecture classes that should give the students a hands-on training and familiarize them with the development environment.

The Laboratory works should enable students to setup and run their own development environment and provide them with the base for their application development and publishing. It should also focus on publishing the application to the App Store and also monetize the apps using one of the advertising networks provided by the SDK.

Text Books:

1. Zigurd Mednieks, Liard Domin, G.Blake Meike, Mausami Nakamura "Programming Android: Java Programming for the New Generation of Mobile Devices", 2nd Edition , O'Reilly 2012
2. Reto Meier, "Professional Android Application Development", Wiley Publishing Inc, 2009, ISBN: 978-0-470-34471-2

Reference Books

1. Barry Burd, Android Application Development All-in-One For Dummies, John Wiley & Sons, Inc 2012.

Course Description

This course in Natural Language Processing is designed to introduce different levels of text analysis and computational aspects of written natural language. It will help students to appreciate complexities involved by focusing on a real world solution to each level of analysis.

Course Objectives

At the end of the course, students are expected to

1. define the nature and characteristics of a language and language processing;
2. apply computational techniques to model and analyze word, syntactic, semantic and pragmatic level of analysis;
3. conduct these processing techniques to design language supported systems.

Course Contents

- | | |
|---|--------------|
| 1. Introduction to NLP | 3 hrs |
| Origins and Importance of NLP | |
| Challenges in NLP (Difficulties, Ambiguities and Evolution) | |
| Language and Knowledge (Phonetics, Syntax, Semantics, Pragmatics and Discourse) | |
| A multi-disciplinary field (Psychology, IR) | |
| Application of NLP | |
| 2. Localization and Language Computing | 4 hrs |
| Introduction to Localization | |
| Key concepts of Localization | |
| Unicode and Multilingual Computing | |
| 3. Finite State Machine (FSM) and Morphology | 6 hrs |
| Introduction to FSM and FST | |
| Morphological Processes | |
| Principles of Word Construction (Suffix, Prefix, Stem, Affixes) | |
| English, Nepali, Turkish | |
| Morphological Representation and FSM | |
| Lexicon, Morphotactics and Orthographic Rules | |
| Morphological parsing and FST | |
| Mealy Machine | |
| FST Operations | |
| 4. Parts of Speech (PoS) Tagging and Hidden Markov Models (HMM) | 4 hrs |
| PoSTagsets | |
| Rule-based PoS tagging | |
| Stochastic PoS tagging | |
| Transformation based tagging | |

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|----|--|-------|
| 5. | Syntactic Analysis
Context Free Grammar(CFG) & Probabilistic CFG
Words' Constituency (Phrase-Level , Sentence-Level)
Parsing (Top-Down and Bottom-Up)
CYK Parser
Probabilistic Parsing | 7 hrs |
| 6. | Lexical Semantics
Lexeme, Lexicon, Senses, Lexical relations
WordNet (Lexical Database)
Word Sense Disambiguation (WSD)
Word Similarity | 7 hrs |
| 7. | Pragmatics & Discourse Analysis
Monologue and Dialogue
Reference Resolution
Coherence and Cohesion
Discourse Structure | 7 hrs |
| 8. | Application of NLP
Question Answering
Machine Translation
Sentiment Analysis
Summary Generation | 7 hrs |

Text Book

Jurafsky, Daniel, James H. Martin. 2000. *Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*. Pearson Education: Singapore.

Reference Books

1. Manning, Christopher D., HinrichSchütze. 1999. *Foundations of statistical natural language processing*. Cambridge: The MIT press.
2. PAN Localization - Guide to Localization of Open Source Software. Nepalinix Team, Madan PuraskarPustakalaya, Nepal. Printworks, Pakistan n.d.
3. Unicode Demystified. A Practical Programmer's Guide to the Encoding Standard. Richard Gillam. Addison-Wesley, 2003.
4. Natural Language Processing with Python. Stephen Bird, Ewan Klein, and Edward Loper. O'Reilly Media, 2009. <http://www.nltk.org/book/>

Course: Data Warehouse and Data Mining

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

- The objective of the course is to understand the underlying principles and techniques of Data Warehousing and Data mining.

Course Contents:

- 1. Introduction (3hrs)**
Concept of Data Mining and Data warehousing, client server and Data Warehouse, DBMS vs DM, DM in Distributed Data bases. Benefit of data mining. Decision Support System
- 2. Data ware Housing and Architecture (9hrs)**
Data warehousing, data Warehouse, Architecture Data warehouse backend process multidimensional data model, OL, AP operators, Warehouse schema and architecture, warehouse server, Meta data, OL, AP engine, KDD Data selection, Data cleaning, Data – Data coding) Data marts.
- 3. Data Mining and Techniques –I (12hrs)**
Data mining architecture, Data Mining tasks, Data mining techniques, Discovery and Analysis of patterns -----, Algorithm for association
- 4. Data Mining Technique – II (15hrs)**
Decision trees, Tree construction principle, Tree construction Algorithm, Pruning techniques, Neural Net, Learning in Neural Net, Unsupervised learning, Data mining using Neural net. Genetic algorithm, Web mining, Content mining, Structure mining, Usage mining, Text mining, Unstructured text, Data mining in E-Commerce
- 5. Data Mining Trends, Issues and Application (6hrs)**
Trends in data mining, Data mining, Application, Security and privacy issues in data mining.

Laboratory Work

The instructor may choose suitable case study to be implemented using data mining.

Reference Books:

- Pujari A.K "Data Mining Techniques "university Press (India) Limited, Hyderabad., India 2091.
- Han Hawel . M Kamber "Data Mining Concepts and Techniques "Academic Press. Hare-----
- Data Ming "Person Education ---
- Sam Anahory, Denris Murray "Data warehousing in the real ----. Person Education

Pokhara University
Faculty of Science and Technology

Course No.: Elective	Full marks: 100
Course title: Generative Artificial Intelligence (3-1-2)	Pass marks: 45
Nature of the course: Theory/Tutorial/Practical	Time per period: 1 hour
Year:	Total periods: 45
Level: Undergraduate	Program: BE Computer/IT/Software

Course Description

This course offers an in-depth introduction to Generative Artificial Intelligence (GenAI), combining foundational programming skills with cutting-edge developments in large language models and AI application frameworks. Students will begin by mastering Python and essential data libraries, then delve into the architecture of LLMs, advanced prompting techniques, and tools like LangChain for building complex generative systems. The course also covers Retrieval Augmented Generation (RAG), ethical considerations in AI, and emerging topics such as multimodal and agentic systems. By the end of the course, students will be equipped to design, build, and critically evaluate GenAI applications across a range of contexts.

General Objectives

The course is designed with the following objectives:

1. Build foundational skills in Python and essential data libraries for AI development.
2. Understand the architecture, training, and evolution of large language models in generative AI.
3. Apply prompt engineering, LangChain, and RAG techniques to build and optimize GenAI applications.
4. Analyze ethical, social, and practical implications of GenAI in real-world contexts.

Methods of Instruction

Contents in detail with specific objectives

Specific Objectives	Contents
Develop foundational Python skills and familiarize with core data libraries and tools useful for generative AI development.	Unit 1: Introduction to Generative AI and Python Fundamentals (7 hrs) 1.1 Evolution of AI and Emergence of Generative AI 1.2 Python Syntax and Data Types 1.3 Control Structures and Functions in Python 1.4 Introduction to NumPy for Numerical Operations 1.5 Pandas for Data Manipulation and Analysis 1.6 Data Visualization with Matplotlib 1.7 Introduction to FastAPI for API development
Familiarize with the architecture, types, and training methodologies of modern large language models.	Unit 2: Foundations of Large Language Models (6 hrs) 2.1 Transformer Architecture and Attention Mechanisms 2.2 Key LLM Types: GPT, BERT, LLaMA, 2.3 Training Methodologies: Pre-training and Fine-tuning
Implement the zero-shot and few shot prompts to build effective Gen AI applications.	Unit 3: Prompt Engineering Fundamentals (8 hrs) 3.1 Introduction to Prompt Engineering 3.2 Anatomy of Effective Prompts 3.3 Zero-shot Learning Techniques 3.4 Few-shot Learning and Examples 3.5 Chain-of-Thought Prompting

	3.6 Role Prompting and System Instructions 3.7 Prompt Templates and Frameworks 3.8 Case Study: Interactive Prompt Workshop with LLM models
Build complex LLM-based applications using the LangChain framework by leveraging chains, memory, and tool integration.	Unit 4: LangChain Framework and Applications (8 hrs) 4.1 LangChain Architecture and Components 4.2 Chains and Sequence Processing 4.3 Prompt Templates and LLM Integration 4.4 Memory Systems in LangChain 4.5 Agents and Tools Implementation 4.6 Document Loading and Text Splitting 4.7 Building Conversational Applications 4.8 Case Studies of LangChain Applications
Implement context-aware LLM systems using vector embeddings, similarity search, and evaluation metrics in RAG pipelines.	Unit 5: Retrieval Augmented Generation (RAG) (8 hrs) 5.1 Introduction to Retrieval Augmented Generation 5.2 Understanding Tokens and Token Limitations 5.3 Vector Embeddings: Principles and Generation 5.4 Vector Databases: Pinecone, Chroma, 5.5 Chunking Strategies and Metadata 5.6 Similarity Search and Retrieval Mechanisms 5.7 Hybrid Search Methods 5.8 Re-ranking Techniques 5.9 Evaluation Metrics: BLEU, Context precision, recall, faithfulness, relevance
Analyze ethical challenges in generative AI and apply responsible AI practices across the model lifecycle.	Unit 6: Ethical and Responsible AI (4 hrs) 6.1 Introduction to Ethics in Generative AI 6.2 Sources and Impacts of Bias in LLMs 6.3 Fairness, Representation, and Social Harm 6.4 Explainability and Transparency Challenges

	6.5 Model Misuse, Hallucinations, and Jailbreak Risks 6.6 Data Privacy, Consent, and Legal Considerations (e.g., GDPR, AI Act)
Familiarize students with emerging trends in generative AI, including multimodal systems, agentic models, and advanced fine-tuning techniques.	Unit 7: Advanced Topics in Generative AI (4 hrs) 7.1 Fundamentals of Multimodal AI 7.2 Agentic AI Systems 7.3 Emerging Research Directions 7.4 Self-hosting LLMs 7.5 Finetuning LLMs

List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course.

1. Python Environment Setup & Basics: Installing Python, Jupyter, and essential libraries; basic syntax and data types practice.
2. NumPy & Pandas Fundamentals: Core operations with NumPy arrays and Pandas DataFrames for AI data preparation.
3. Data Visualization & FastAPI: Creating visualizations with Matplotlib and building simple APIs with FastAPI.
4. LLM API Integration: Making calls to popular LLM APIs and handling responses.
5. Zero-shot Prompting: Crafting and testing effective zero-shot prompts across various tasks.
6. Few-shot Prompting: Implementing few-shot learning techniques with examples.
7. Chain-of-Thought Prompting: Developing prompts for complex reasoning tasks.
8. Role Prompting & System Instructions: Creating specialized prompts for different contexts and roles.
9. Prompt Templates & Frameworks: Building reusable prompt structures and formats.
10. LangChain Basics: Setting up LangChain environment and implementing basic chains.
11. LangChain Memory Systems: Implementing different memory types for contextual conversations.
12. LangChain Agents & Tools: Building agents that can use tools to solve multi-step problems.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory				
Attendance & Class Participation				
Assignments				
Presentations/Quizzes				
Internal Assessment				
Practical				
Attendance & Class Participation				
Lab Report/Project Report				
Practical Exam/Project Work				
Viva				
Total Internal				

Student requirements:

Pokhara University
Faculty of Science and Technology

Course No.:

Full marks: 100

Course title: Information System Audit(3-1-0)

Pass marks: 45

Nature of the course: Theory

Total Lectures: 45 hrs.

Level: Bachelors

Program: BE

1. Course Description

The Information System Audit course for undergraduate students provides an in-depth understanding of auditing principles, methodologies, and best practices for assessing the security, integrity, and compliance of information systems. The course covers risk assessment, control frameworks, IT governance, regulatory compliance, and emerging threats in cybersecurity. Students will gain hands-on experience in auditing IT infrastructure, applications, and processes through case studies and practical exercises. This course covers different concepts of Information Systems Auditing including basics, hardware and software security issues, information systems audit and conducting IS audit, risk-based systems audit, business continuity and disaster, auditing in the ICT environment, and security testing.

2. General Objectives

- To develop a foundational understanding of Information systems auditing principles, processes, and methodologies.
- To explore risk-based auditing approaches within ICT environments.
- To examine the role of Governance, Risk, and Compliance (GRC) in Information systems auditing.
- To understand the importance of business continuity and disaster recovery in audit planning.
- To gain familiarity with security testing tools, vulnerability assessment, and penetration testing techniques.
- To analyze emerging trends and challenges in the field of Information systems auditing.

3. Methods of Instruction

Lecture, Discussion, Readings, Case Studies and Group Workshops

4. Contents in Detail.

Specific Objectives	Contents
Acquire knowledge of the fundamental concepts of Information System Audit and IT Audit Frameworks, with an emphasis on widely recognized standards such as ISO 27001, NIST, and MITRE.	Unit 1: Overview of Systems Audit (8 Hrs.) 1.1 Information Systems Audit and Information Systems Auditor 1.2 Legal Requirements of an Information Systems Audit 1.3 Systems Environment and Information Systems Audit 1.4 Information Systems Assets and Classification of Controls 1.5 Information Systems Audit Coverage 1.6 IT Audit Framework, ISO 27001, NIST Cyber Security Framework and MITRE
Gain knowledge of the key concepts related to hardware and software security issues encountered during an Information System Audit.	Unit 2: Hardware and Software Security Issues (8 Hrs.) 2.1 Hardware Security Objective 2.2 Peripheral Devices and Storage Media 2.3 Authentication Devices 2.4 Hardware Acquisition, Hardware Maintenance and Management of Obsolescence 2.5 Disposal of Equipment; Problem Management; Change Management 2.6 Network and Communication Issues. 2.7 Overview of Types of Software; Elements of Software Security 2.8 Control Issues during Installation and Maintenance
Gain knowledge of the requirements of Information System Audit in relation to system controls, log management, and evidence collection processes.	Unit 3: Information Systems Audit Requirements (8 Hrs.) 3.1 Information Systems Control Objectives; Information Systems Audit Objectives; 3.2 System Effectiveness and Efficiency 3.3 Information Systems Abuse 3.4 Asset Safeguarding Objective and Process 3.5 Evidence Collection and Evaluation 3.6 Logs and Audit Trails as Evidence 3.7 IT Audit Standard and Regulatory Requirements
Gain knowledge of the concept of Information System Auditing, including its purpose, scope, and role in evaluating and ensuring the effectiveness of IT controls and processes.	Unit 4: Conducting an Information Systems Audit (10 Hrs.) 4.1 Audit Program and Audit Plan 4.2 Audit Procedures and Approaches 4.3 System Understanding and Review 4.4 Compliance Reviews and Tests 4.5 Substantive Reviews and Tests 4.6 Audit Tools and Techniques 4.7 Sampling Techniques 4.8 Audit Questionnaire; Audit Documentation; Audit Report 4.9 Auditing Approaches; Sample Audit Work-Planning Memo

	4.10 Sample Audit Work Process Flow 4.11 Conducting a Risk-Based Information Systems Audit 4.12 Risk Assessment and Risk Management Strategy
Gain knowledge of the concept of Business Continuity and Disaster Recovery Planning within the context of Information System Audits, focusing on strategies to ensure organizational resilience and recovery in the event of disruptions.	Unit 5: Business Continuity and Disaster Recovery Plan (6 Hrs.) 5.1 Business Continuity and Disaster Recovery Process 5.2 Business Impact Analysis; Incident Response Plan 5.3 Disaster Recovery Plan 5.4 Types of Disaster Recovery Plans 5.5 Emergency Preparedness Audit Checklist 5.6 Business Continuity Strategies 5.7 Business Resumption Plan Audit Checklist 5.8 Recovery Procedures Testing Checklist; Plan Maintenance Checklist.
Apply the concept of VAPT and Emerging Concepts	Unit 6: Security Testing and Cloud Computing Audit (5 Hrs.) 6.1 Cybersecurity 6.2 Vulnerability Assessment and Penetration Testing (VAPT) 6.3 Security Testing Tools 6.4 Emerging Concepts in Information System Audit

5. Tutorial and Group Workshops(15 Hours)

The tutorial and group workshop should cover the following works:

SN	Group Workshops
1.	Information Security Gap Assessment through ISO 27001
2.	Cybersecurity Maturity Assessment through NIST Cybersecurity Framework using CISA-CSET or similar tools
3.	Conduct Information System Audit and Reporting
4.	Conducting Risk Assessment
5	Case Study: GRC Cybersecurity Engine

6. Evaluation system and Students' Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment and group workshop. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory			Semester-End examination	50
Attendance & Class Participation	10%	5		
Assignments	10%	5		
Presentations/Quizzes	10%	5		
Internal Assessment	50%	25		
Group Workshops	20%	10		
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

References:

1. Hall, J. A. (2020). *Information Technology Auditing and Assurance* (5th ed.). Cengage Learning
2. Moeller, R. (2022). *IT Audit, Control, and Security* (3rd ed.). Wiley.
3. ISACA. (2024). *CISA Review Manual*. ISACA.
4. ISACA. (2022). *COBIT 2019 Framework: Governance and Management of Enterprise IT*. ISACA
5. Veena Hingarh and Arif Ahmed (2013), *Understanding and Conducting Information Systems Auditing*, Wiley
6. Jack J. Champlain (2003), *Auditing Information Systems*, 2nd Edition, Wiley
7. Richard Cascarino (2007), *Auditor's Guide to Information Systems Auditing*, Wiley
8. ISACA Journals
9. ISACA IT Audit Frameworks
10. NIST Special Publications

Advanced Web Technology

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	100	20	100

Course Objectives: To improve basic understanding of internet. The course also aims to build foundation on HTML, Java Script, VB Script, ASP, DHTML, and CGI-perl

Contents:

1. Introduction to the World Wide Web

[5hrs]

Connecting to the internet: Dial up connection, Direct Connection, Internet Protocol: TCP/IP, FTP, Telnet, Gopher, WAIS, Internet Addressing: IP address, Domain Name, Electronic mail address, Uniform Resource Locator, Web page and HTML, Introduction to the Internet Browser, Searching on the Web: Web Index, Search Index, Meta search Engine, Internet Chat.

2. Introduction to the HTML

[8hrs]

Different section of an HTML, Common tags, Text Style, Linking, Images, Formatting text with , Special Character, Horizontal Rulers and line Breaks, unordered list, Nested and ordered list in HTML, Tables, complex tables with formatted and color, simple form including text area, Tables, complex tables with formatted and color, simple form including text area, password boxes, and check boxes, radio button, and pull down boxes, frames.

3. Dynamic HTML

[5hrs]

DHTML vs HTML, Introduction to the Cascading Style Sheets, Declaring style in the header section of the document, linking external style sheet.

4. Introduction to the Java Script and VBScript

[12hrs]

Java Script

Introduction, Accepting input from the users, Basic arithmetic operators, Sequence Structure, Repetition structure, introduction to function, Array, Objects: Math, String, Data

VBScript

Introduction, Operators, Data Type, Control statements, Vbscript Function, Array, String Manipulation, Class and Object, Using Vbscript code to respond to an events.

5. Active Server Page and CGI-Perl

[12hrs]

Active Server Page

Introduction, Simple ASP Document, Client Side request, Data base connectivity, Server Side Active-X components, Simple Example: Connectivity to a database, User log into a site, Validate User Login

CGI-Perl

Introduction the Perl, String processing and regular Expression, viewing client/server environment variable, Form processing and Business logic, server side includes, varifying a username and Password

6. E-commerce

[3hrs]

Taking charge: buying cars, Books, unusual item, shopping for supplier, online shopping Risk and safe guards, Secure server and secure Web pages, commercial sites and self regulation, online Auction, National security and Hackers, digital Telephony and Covergence, Internet Appliances.

Laboratory Exercise:

At least Five labs for each Module mentioned in the syllabus and at the end of the semester, small project including development of the Web site which should cover entire syllabus.

Reference Books:

1. Internet and World Wide Web: How to Program by Dietel & Nieto
2. Fundamental of Information Technology: Alexis Leon & Mathew Leon
3. The ABC of java Script: Lee Purcell & Mary Jane Mara
4. Web 101: Making the Network for you: Wendy Lehnert
5. Teach Yourself Web Technology: I van Bayross: Part I and II

Routing and Switching:

Course Objective:

Understand the ipv4 addressing and subnetting, configure and understand various routing protocols: static, dynamic (Distance vector and link state) and switching advance topics: STP and VLAN.

1 IP addressing and Subnetting

3hrs.

Pv4 addressing, Classes of IP (A,B,C,D and E), Reserved IP addresses, Public and Private IP address, IPv4 vs Ipv6, Need for subnetting, Binary method of subnetting, Classless and Classful addressing (CIDR, Supernetting or IP summarization), Subnet in your Heads, VLSM,

2 Cabling

1hr

Coaxial cable , STP & UTP cable, UTP cabling (straight through & cross over) CAT5 and CAT6, Optical fiber, Serial cable and Rollover cable for router

3 Introduction to Routers

3hrs

Routers function, Internal configuration component, Modular and non Modular Router, Routers physical interfaces ,Management ports (console port and Auxiliary port), Console port connections, Hyperterminal (windows), minicom(console based) or cutecom(Graphical) in Linux ,Connecting LAN Interfaces, Connecting WAN Interfaces

4 Operating Cisco Router

2hrs

The purpose of Cisco IOS software, IOS naming convention, Three distinct operating environments or modes (Roman, boot ROM, normal IOS), Steps in Router initialization, Router user interface modes (User exec and privileged or enable exec), Other router modes global mode and specific modes, Keyboard help in router CLI (?, tab), Enhanced Editing commands (like ctrl -A, Esc -B, Esc -F, Ctrl-E), Show version and show flash commands ,Show history and setting the clock

5 Configuring the Router

3hrs

Configuring a router name, Configuring router passwords, Console password, Vty password, Enable password, Perform password encryption, Examining show commands, Configuring a serial interface, Configuring Ethernet interface, Making configuration changes and saving the configuration, Configuring interface description, Login Banners, Host name resolution and configuring host tables

6 Routing and Routing protocols:

6hrs

Difference between routing and routed protocol, Packet propagation and switching within the router, Path determination, Routing tables, Routing algorithms and metrics, Static and Dynamic Routing protocol, Administrative distance, Default Routing, Autonomous system, IGP and EGP, Distance vector and Link state Routing protocols, Hybrid Routing protocol, Difference between distance vector and link state routing protocols, Routing loops and its removal techniques, Routing Information Protocol(RIP), RIP routing process and its characteristics, Configuring RIP, Ip classless command, RIP version 1 & version 2, Verifying RIP configuration, Troubleshooting RIP updates issues, debug command, Preventing Routing updates through an interface, the passive interface, RIP load Balancing, Integrating static routes with RIP, backup route

7 Enhanced Interior Gateway Routing Protocol(EIGRP)

5hrs

Introduction to classless routing protocol, Difference between EIGRP & IGRP, EIGRP Features, EIGRP Terminology, Neighbor Table, Topology Table, Routing Table, Successor, Feasible Successor, Fields in topology table, Feasible distance, Route source, Reported distance, Interface information, Route status, Reliable Transport Protocol (RTP), Diffusing Update Algorithm (DUAL), EIGRP packets, Configuring EIGRP, Configuring Dis contiguous Networks, IP summarization and Manual IP, summarization in EIGRP, Verifying EIGRP

8 Open Shortest Path First (OSPF)

6hrs

Single Area OSPF concepts, OSPF characteristics, OSPF Terminology, Adjacency Database, Link and Link-state, Topological Database (or Link StateDatabase), Area and cost, Routing Table, Designated router(DR) and Backup Designated Router(BDR), OSPF network Types, Broadcast Multiaccess point-to-point,NBMA, Point-to-multipoint, OSPF Hello protocol, OSPF configurations, Wildcard mask, Configuring OSPF loopback address and router priority, DR and BDR Election, Configuring OSPF authentication, Simple password Authentication, Message Digest Authentication (Md5), verifying OSPF

9 Access Control List (ACL)

3hrs

Introduction to ACL, Inbound and Outbound Interfaces, Types of ACL, Standard ACL, Extended ACL, Named ACL, Configuring standard ACL, Configuring Extended ACL, Configuring Named ACL, Verifying ACLs, Restricting Virtual Terminal Access

10 Switching concepts and Configuration

4hrs

Ethernet switching and CAM Table, Microsegmentation, Switches and Collision domain, Switches and Broadcast domain, Layer 2 and Layer3 switching, Symmetric and Asymmetric switching, LAN switch Types, Store and Forward, Cut-through (FastForward), Fragment Free (modified cut-through), Switch LDE indicators, Configuring the IP address, hostname

and passwords , Managing the MAC address Table , Configuring the static MAC address table , Configuring the port security , Password Recovery ,IOS backup and Upgrade , Backup configuration file

11 Spanning Tree Protocol (STP)

2hrs

Redundant Switch Topology & Bridging loops , Broadcast Storms, Multiple frame copies , MAC address table instability problem , Preventing Loops with Spanning Tree Protocol ,Spanning Tree communication:Bridge Protocol Data Unit , Spanning Tree operation, Electing Root Bridge , Electing Root ports , Electing Designated ports, STP states : Disabled ,Blocking ,Listening ,Learning ,Forwarding

12 Virtual LANS (VLANs) and Trunking

4hrs

VLAN Introduction , Broadcast Domain with VLAN and router , VLAN membership: Static and, Dynamic VLAN , Benefits of VLAN , Configuring static VLAN , Verifying and Deleting VLANs , Access link & Trunk Link , VLAN Identification , Inter-switch Link Protocol (ISL) , IEEE 802.1Q ,Trunking Overview , Configuring a VLAN Trunk , Removing VLANs from a trunk , VLAN Trunking Protocol (VTP) , VTP Domains , VTP modes , VTP configuration , Configuring a VTP management domain , Configuring the VTP mode , configuring the VTP password , Configuration revision number Inter-VLAN Routing , Sub-Interfaces in Router , Router on a stick ,VTP Pruning

13 Scaling IP addressing :

3hrs

Scaling IP address with NAT and PAT , NAT overview , NAT Terminology , Inside local address , Inside global address , Outside local address , Outside global address , Dynamic NAT , Dynamic NAT, configuration , Dynamic NAT configuration example , Static NAT , Static NAT , NAT overload or PAT ,Configuring NAT overload , Verifying NAT translations , Troubleshooting NAT translation , NAT consideration , NAT advantages , NAT disadvantages .

PJJ 430.3 Programming in Java

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Contents:

1. Elements Of Java Language (3 Hrs)

Java as a Programming tools, Benefits of Java, Historical Background of Java, A simple Java Program. Data type, Variable, Assignment and Initialization, Operator, String, Control Flow, Class Method (User Defined Function), Array.

2. Object Oriented Programming In Java (9 Hrs)

Introduction to object oriented programming in Java, Reusability using Existing classes, Building User defined class, Package, Inheritance, Casting Abstract classes, Access Protection Mechanism, Reflection, Designing Inheritance , Interface, Inner Classes.

3. Exception, Stream and I/O (3 Hrs)

Handling Error and Exception, catching Exception, tips on handling Exception, Debugging techniques. Stream, Zip files Stream, Object Stream, and Handling Files.

4. Applets and Application (4 Hrs)

Fundamental concept of Applet, Simple Applet, Testing Applets, Converting Application to Applets, Applets HTML tags and Attribute. Pop -UP Windows in Applet, Multimedia Applets context.

5. Events, Handling Events and AWT/Swing (6 Hrs)

Basic of Event handling, AWT Event hierarchy, semantics and low level Events in AWT, Event Handling, Individual Events. Separating GUI and Application code, Multicasting, Advance Event Handling. An Introduction of layout management, Text input choice, scroll Bar, complex layout management, Menus, Dialog Box.

6. Graphics and Images / Animation / Multimedia (5 Hrs)

Introduction to Graphics Programming, creating Closable frames. Terminating graphics program. Frame layout displaying information in a frame. Graphics object. Text and fonts, color. Drawing shapes from lines drawing rectangle and Ovals. Filling shapes paint mode images.

7. Network Programming

(8 Hrs)

Networking Basics, introduction to Socket, Socket Programming, Understanding Port, Networking Classes in Java, Creating Own Server and Client in Java, Creating Multithread Java Server, URL and URL connection Class.

8. Java Database Connectivity (JDBC)

(7 Hrs)

Understanding JDBC, database Driver, JDBC-ODBC Bridge, Java Native Driver, Intermediate Database Access Server, JDBC API, Making a JDBC Application, Using Prepared Statement.

Reference Books:

1. Dietel H.M and Dietel P.J., Java: How to Program, Third Edition, Pearson Education Asia
2. Naughton Java 2: The Complete Reference, Tata McGraw Hill
3. Balagurusamy E., Programming in Java: 2nd Edition, Tata McGraw Hill

Information Retrieval (3 – 0 – 0)**(Elective Course)****Evaluation:**

	Theory	Practical	Total
Sessional	50	-	50
Final	50	-	50
Total	100	-	100

Course Description:

The course is intended to prepare the students to understand the underlying theories and algorithms of text based information retrieval systems and to introduce the methodology for the design and evaluation of information retrieval (IR) systems. The core components of the course include statistical characteristics of text, representation of information needs and documents, several important retrieval models and experimental evaluation. The course will focus on the both the theoretical aspects of information retrieval design and evaluation, and will also consider the practical aspects of how these theories have been implemented in actual systems.

Course Objectives:

The main objectives of the course are:

1. To provide the basic understanding of an information retrieval system
2. To introduce the core components of a modern information retrieval system
3. To familiarise with the working of search engines
4. To learn to evaluate the performance of an information retrieval system

Course contents:

1. **Information Retrieval History and Key Concepts (4 hrs)**
Preliminaries of IR: Information retrieval, document and corpus statistics, tokens and stopwords, stemming, lemmatisation, information need, query and search, indexing, relevance, evaluation; Related areas to IR; the impact of the web on IR
2. **Boolean Retrieval (4 hrs)**
Boolean retrieval model: Term-document matrix, lexicon dictionary, term frequency and document frequency, inverted index, processing boolean queries; query optimisation; similarity measures: cosine similarity, Dice's coefficient, Jaccard's coefficient, overlap coefficient
3. **Vector Space Model (6 hrs)**
Term frequency and inverse document frequency (tf-idf), term-document matrix, bag of words, Zipf's law; vector space, Euclidean distance, dot product and cosine similarity, building queries and computing vector scores, normalisation
4. **Evaluation in Information Retrieval (5 hrs)**
Preliminaries: gold standard, measure of relevance; information retrieval experiments: Text Retrieval Conference (TREC), NII Test Collections for IR Systems (NTCIR) and Conference and Labs of the Evaluation Forum (CLEF); evaluation: precision, recall, ROC curve, accuracy, F-measure, Mean Average Precision (MAP) and Mean Reciprocal Rank (MRR), kappa statistics

5. **Probabilistic Information Retrieval** (6 hrs)
Review of basic probability theory: random variable, chain rule, Bayes' rule, prior and posterior probability and odds of an event; Basics of probabilistic IR: the probability ranking principle (PRP), binary independence model (BIM), Okapi weighting; Bayesian approach to information retrieval
6. **Language Models** (10 hrs)
Preliminaries: finite automata, language; language models (LM): unigram and bigram LM, multinomial distribution, query likelihood model, interpolation and smoothing; extended LM approaches: Kullback-Leibler divergence and translation models
7. **Web Search** (5 hrs)
Preliminaries: introduction and web history, web as a graph, users' need and user experience in web search; components of a large scale web search engine; pagerank, Hypertext-induced text selection (HITS); IR in social networks
8. **Natural Language Processing (NLP) and Information Retrieval** (5 hrs)
Language basics: morphology, parts of speech, grammar, semantics, pragmatics and discourse; word senses and sense disambiguation; indexing by base forms of words, by phrases, by word senses; named entities, anaphora resolution, query expansion; question answering

Text Book:

Manning, Christopher D., Prabhakar Raghavan, and Hinrich Schütze. Introduction to information retrieval. Vol. 1. Cambridge: Cambridge University Press, 2008.

References:

- Baeza-Yates, Ricardo, and Berthier Ribeiro-Neto. Modern Information Retrieval, Addison Wesley Longman, 1999.
- Grossman, D. A., O. Frieder. Information Retrieval: Algorithms and Heuristics. Springer, 2004.
- Moffat, A. A., and Timothy C. Bell. Managing gigabytes: compressing and indexing documents and images. Morgan Kaufmann, 1999.
- Langville, Amy and Carl Meyer. Google's PageRank and Beyond: the Science of Search Engine Rankings. Princeton University Press, 2006.
- Salton, Gerard, and M. J. McGill. Introduction to Modern Information Retrieval. McGraw-Hill, 1983.